

2	(a)	work done = $F \times \Delta x$, where Δx is the distance moved by the object in a unit time, $\frac{\text{work done}}{\text{time taken}} = \frac{F \times \Delta x}{\text{time taken}}, \text{ but } \frac{\Delta x}{\text{time taken}} = \text{velocity}$ So $P = Fv$	B1
			A0
	(b) (i)	at maximum speed, driving force = resistive force (BOD for net force = 0)	C1
		$\frac{P}{v_{\max}} = \text{resistive force}$	A1
		$\frac{200000}{\text{maximum speed}} = 5000$	
		maximum speed = 40 m s^{-1}	
	(ii)	$\frac{P}{v_{\max}} = \text{resistive force} + mg \sin \theta$	C1
		$\frac{200000}{\text{maximum speed}} = 5000 + 1500 \times 9.81 \times \sin 10^\circ$	M1
		maximum speed = 26 m s^{-1}	A1
	(iii)	From chemical potential energy to gravitational potential energy and thermal energy (BOD: work done due against friction)	B1
			B1
	(c)	normal contact force – weight provides centripetal force for contact normal contact force > 0	B1
			B1
		$mg - N = m \frac{v^2}{r}$	
		$N = mg - m \frac{v^2}{r}$	
		When $N > 0$	
		$mg - m \frac{v^2}{r} > 0$	
		$v < \sqrt{rg}$	
		$= \sqrt{250 \times 9.81}$	
		$= 49.5$	B1
		maximum speed = 49 m s^{-1}	A1
		(accept both 49 and 50)	